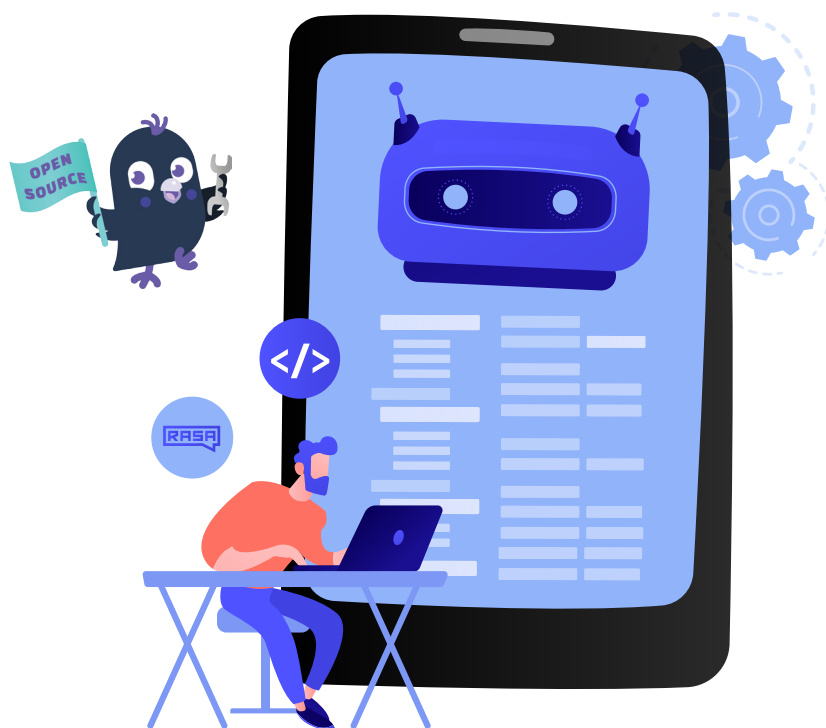


Using the Rasa Framework for Creating Chatbots



“AI (artificial intelligence) is going to change the world more than anything in the history of mankind. More than electricity,” said Dr Kai Fu Lee, AI oracle and venture capitalist. His words ring true today, as AI pervades almost every part of human life. This article talks about how chatbots can be built using the Rasa framework.

All of us use artificial intelligence and machine learning (AIML) technology in some way or the other today — be it video predictions, online shopping, social media, surveillance, food delivery, transportation, self-driving cars, and so on.

Natural language processing (NLP) is a discipline of AI that is used to help understand, interpret and manipulate human language. Let us first look at NLP and its applications, and then build a chatbot from scratch using the Rasa framework.

Natural language processing

NLP helps computers communicate with humans in their own language and scales other language-related tasks. For example, it opens up the possibility for computers to read text, hear and interpret speech, as well as measure sentiment and determine important parts of it.

Today’s machines can analyse more language based data than humans, without fatigue and in a consistent, unbiased way. Considering the staggering amount of unstructured

data, from medical records to social media, that’s generated every day, automation is critical to fully analyse text and speech data efficiently.

Sample applications of NLP include sentiment analysis, customer reviews, customer segmentation, anomaly detection, product improvement; topic modelling such as coming up with new topics from the text and using them to assign new supervised learning labels, providing insights that are too difficult to find from manual searching; and text categorisation of animal species, fake news, bank transactions, etc.

Chatbots

A chatbot is a computer program that simulates human conversation through voice commands, text chats or both. Short for chatterbot, it is an AI feature that can be embedded and used through any major messaging/text application. This automated program interacts like a human, and costs little to nothing to engage with. Key examples are those used by businesses in Facebook Messenger, or as virtual assistants, such as Amazon’s Alexa.